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DC – AI – Routing – Switching – Security – Wireless – Automation

Introducing the ACX7024X

An HPE Juniper Networking Techpost

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1 Introducing the ACX7024X

Nicolas Fevrier - 04/24/2024

1.1 Introduction

We launched the ACX7024 in mid-2022, with the following characteristics:

- 1RU Ethernet router.
- Built for the aggregation of SFP interfaces (1GE/10GE/25GE) to QSFP28 interfaces (40GE/100GE).
- 360Gbps of forwarding capacity powered by a Broadcom DNX Qumran2u.
- Class-D capable based on Juniper's PTP FPGA.
- Hardened device, categorized as i-temp (supports temperatures from -40 to +65C).
- AC or DC power modules, with 1+1 redundancy.
- 6x fixed fans (front to back cooling), 5+1 redundancy.

It was targeting markets like metro and aggregation where routing scale is not a key parameter. In these roles, high volumes of routers are deployed and cost pressure is very important.

That's why it was designed with these constraints in mind, and that reflected in the choice of the CPU and the memory size:

- Intel ATOM Deveron C3508 : 4 cores @ 1.6GHz, single-threaded
- 16GB DRAM

```
CPU INFO:
 ========
 Model name : Intel(R) Atom(TM) CPU C3508 @ 1.60GHz
 Configured CPU Speed : 1600.012 1600.019 1600.005 1600.049 MHz
 Physical cores : 4
 Hyperthreaded cores : 4
 L1 Cache size : 224 kB
 L2 Cache size : 8192 kB
 L23Cache size :
 CPU PAIRS:
 =========
 Core 0 :
 Core 1 : 3
 Core 2 : 0
 Core 3 :
 SYSTEM MEMORY USAGE (MB):
 ========================
 MemTotal: 15995356 kB
```

Consequently, certain Qumran2u scale "maximum dimensions" can't be reached, the CPU memory being the first bottleneck. Several features can't be enabled

- IPFIX
- FIB compression
- RIB sharding
- 3rd party App Hosting. Note: PAA (Paragon Active Assurance) is an exception to that rule and is supported on ACX7024.

And this CPU and DRAM are the key differences between ACX7024 and ACX7024X.

1.2 Why the ACX7024X?



Figure 1: ACX7024X rendering

The ACX7024 (non-X) port density and form-factor happen to be very attractive for “light multi-service edge router” use-cases, particularly for enterprise and small CPEs. In these network roles, hardware hardening (i-temp) support is not required, the routers being operated in DC controlled environments. In the other hand, the support of features like sampling and potentially full IPv4 and IPv6 internet tables is required.

That’s the main motivation behind the introduction of an ACX7024X that only differs from its predecessor at the CPU and DRAM level: same port density and forwarding capability, same ASIC, but a more powerful CPU:

- CPU:
  - Intel Atom Processor C Series (Denverton) C3758R
  - 8 cores (not hyperthread: 8 threads)
  - 2.4GHz

> More info: https://www.intel.com/content/www/us/en/products/sku/204840/intel-atom-processor-c3758r-16m-cache-2-40-ghz/specifications.html

- DRAM: 2x 32GB

```
CPU INFO:
 ========
 Model name : Intel(R) Atom(TM) CPU C3758R @ 2.40GHz
 Configured CPU Speed : 2400.014 2400.000 2399.977 2400.000 2399.998 2400.000 2399.995 2400.025
     MHz
 Physical cores : 8
 Hyperthreaded cores : 8
 L1 Cache size : 448 kB
 L2 Cache size : 16384 kB
 L23Cache size :

 CPU PAIRS:
 =========
 Core 0 : 0
 Core 1 : 5 6 7
 Core 2 : 1
 Core 3 :
 Core 4 : 2
 Core 5 :
 Core 6 : 3
 Core 7 :

 SYSTEM MEMORY USAGE (MB):
 ========================
 MemTotal: 65213484 kB
 MemFree: 55898420 kB
 MemAvailable: 60426500 kB
 Buffers: 483044 kB
 Cached: 4743236 kB
 SwapCached: 0 kB
 SwapTotal: 1048572 kB
 SwapFree: 1048572 kB
```

With this new control plane, the router supports commercial c-temp environments, and no longer i-temp.

Again, the same distribution, and port capabilities/configuration is available on ACX7024X:

```
regress@acx7024x-01> show chassis hardware detail
Hardware inventory:
Item Version Part number Serial number Description
Chassis GC1723ANxxxx JNP7024X [ACX7024X]
PSM 0 REV 03 740-134837 1F35D03xxxx JPSU-400W-AC-AFI
PSM 1 REV 03 740-134837 1F35D03xxxx JPSU-400W-AC-AFI
Routing Engine 0 REV 01 650-170107 GC1723ANxxxx RE-ACX-7024X
  mmcblk0 38000 MB MMC 1414xxxx eMMC Disk
CB 0 BUILTIN BUILTIN Control Board
FPC 0 BUILTIN BUILTIN ACX7024X-FPC
  PIC 0 BUILTIN BUILTIN MRATE- 24xSFP28 + 4xQSFP
    Xcvr 0 REV 01 740-067443 QI29xxxx QSFP+-40G-SR4
    Xcvr 1 REV 01 740-061405 1ECQ14xxxx QSFP-100GBASE-SR4-T2
    Xcvr 2 REV 01 740-032986 QD51xxxx QSFP+-40G-SR4
    Xcvr 3 REV 01 740-061405 1ECQ132xxxx QSFP-100GBASE-SR4-T2
    Xcvr 4 REV 01 740-021308 N9Lxxxx SFP+-10G-SR
    Xcvr 5 REV 01 740-021308 N9Lxxxx SFP+-10G-SR
    Xcvr 6 REV 01 740-021308 N9Lxxxx SFP+-10G-SR
    Xcvr 7 REV 01 740-021308 N9Kxxxx SFP+-10G-SR
    Xcvr 8 REV 01 740-021308 N9Lxxxx SFP+-10G-SR
    Xcvr 10 REV 01 740-021308 MTKxxxx SFP+-10G-SR
    Xcvr 13 REV 01 740-031980 A2Pxxxx SFP+-10G-SR
    Xcvr 14 REV 01 740-031980 A2Pxxxx SFP+-10G-SR
Fan Tray 0 BUILTIN BUILTIN ACX7024X Fan, Front to Back Airflow - AFO
regress@ acx7024x-01>
```

and

```
regress@acx7024x-01> show chassis pic fpc-slot 0 pic-slot 0
FPC slot 0, PIC slot 0 information:
  Type MRATE- 24xSFP28 + 4xQSFP
  State Online
  PIC version 00.00
  Uptime 1 day, 10 hours, 17 minutes, 18 seconds
<SNIP>
Port speed information:
  Port PFE Capable Port Speeds
  0 0 1x100G 1x40G 2x50G 4x25G 4x10G
  1 0 1x100G 1x40G 2x50G 4x25G 4x10G
  2 0 1x100G 1x40G 2x50G 4x25G 4x10G
  3 0 1x100G 1x40G 2x50G 4x25G 4x10G
  4 0 1x10G 1x1G 1x25G
  5 0 1x10G 1x1G 1x25G
  6 0 1x10G 1x1G 1x25G
  7 0 1x10G 1x1G 1x25G
<SNIP>
  25 0 1x10G 1x1G 1x25G
  26 0 1x10G 1x1G 1x25G
  27 0 1x10G 1x1G 1x25G
  28 NA 1x10G
regress@acx7024x-01>
```

You can refer to the previous techpost article from Pankaj on ACX7024 for more details: https://techpost.hpe.com/articles/acx7024-deepdive/article

1.2.1 Other Differences?

With new CPU and larger DRAM, we expect less than 20W of power usage difference between the ACX7024X and the ACX7024.

1.3 Benefits

On ACX7024X, we now support the four different MDB profiles.

```
regress@acx7024x-01# set system packet-forwarding-options hw-db-profile ?
Possible completions:
+ apply-groups Groups from which to inherit configuration data
+ apply-groups-except Don't inherit configuration data from these groups
  bng Selects BNG DB profile, restarts PFE
  carrier-ethernet Selects Carrier Ethernet high scale MAC DB profile, restarts PFE
  cloud-metro Selects Cloud-Metro DB profile, restarts PFE
  lean-edge Selects Lean Edge max IPv4/IPv6 FIB DB profile, restarts PFE
[edit]
regress@ acx7024x-01#
```

By default, it will boot up in lean-edge mode and in the example below, the router receives a full internet view from an EBGP peer. That's almost 1 million IPv4 entries and more than 200,000 IPv6.

```
regress@rtme-acx7024x-01:pfe> show route summary
IPv4 Route Tables:
Index Routes Size(b) Prefixes Aggr Installed Comp(%) Errors
-------- ---------- ---------- --------- --------- ---------- ------ -------
Default 976921 181338360 954934 534232 170237 83 0
1 0 0 0 0 0 0 0
50 5 600 5 0 5 0 0
36738 5 600 5 0 5 0 0
<...>
IPv6 Route Tables:
Index Routes Size(b) Prefixes Aggr Installed Comp(%) Errors
-------- ---------- ---------- --------- --------- ---------- ------ -------
Default 207095 35579400 204010 89400 55788 73 0
1 0 0 0 0 0 0 0
50 6 720 6 0 6 0 0
36738 6 720 6 0 6 0 0
<...>
regress@rtme-acx7024x-01:pfe>
```

It's of course an ideal case, since we have a single BGP neighbor. But still you can see the FIB compression in action with a ratio of 83% for IPv4 and 73% for IPv6.

In terms of resource occupation, that's the appid 12 here with 212k IPv4 and 66k IPv6 entries for a total 346k used out of the 1.18M capacity.

```
regress@rtme-acx7024x-01:pfe> show evo-pfemand resource appdb appid 12
UNIT 0
App LPM-v4 LPM-v6
-----------------------------------------------------
Route 213294 66579
regress@rtme-acx7024x-01:pfe>

regress@rtme-acx7024x-01:pfe> show evo-pfemand resource terse-usage
UNIT 0 MDB Profile: custom_lean_edge_jnpr
Resource Usage Capacity size
COS_HR_ELEM 75 512 1
COS_VOQ 600 32768 1
COS_VOQ_CNCTR 600 49152 1
ECMP 3 32000 30
EEDB_7 4 16384 0
EEDB_8 43 16384 0
EGR_VSI 8 32768 30
FAILOVER 145 6000 30
FEC_1 4 32768 150
FEC_3 13 32768 150
IFL_STATS 4 8000 1
ING_MC_GRP 5 262144 1
ING_VSI 8 21846 90
INLIF_1 39 65536 60
ISEM_1 8 65536 30
KAPS 346452 1184000 0
LEM 2 393216 30
TCAM_IPMF1_80 560 0 1
TCAM_IPMF3_80 6 0 1
TCAM_PMF_ALL_INTRN_80566 26624 1
regress@rtme-acx7024x-01:pfe>
```

So, we have plenty of room to grow.

Other features disabled on ACX7024 can now be used, like software jFlow and IPFIX.

1.4 Conclusion

The ACX7024X is a very competitive 360Gbps router with 24x SFP, 1GE, 10GE or 25GE and 4 ports QSFP 40GE or 100GE. PTP class-D capable and with a large DRAM space for CPU-intensive features. It supports a full internet table (with both IPv4 and IPv6 routes) and can be used in multiple enterprise roles or for small-size CPEs.

1.5 Useful links

- ACX7024 Deepdive article: https://techpost.hpe.com/articles/acx7024-deepdive/article

1.6 Glossary

- CPE: Customer Premises Equipment (router)
- CPU: Compute Processor Unit
- DRAM: Dynamic Random Access Memory
- FIB: Forwarding Information Base
- PTP: Precision Time protocol
- QSFP: Quad Small Form-Factor Pluggable
- RIB: Routing Information Base
- SFP: Small Form-Factor Pluggable

1.7 Acknowledgements

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